

MAJORS & MINORS

Pre-Engineering Pathway

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EMAIL

CALL



Engineering is a creative application of scientific principles, focused on designing or developing structures, machines, or manufacturing processes that realize a desired objective.

Southwestern’s Pre-Engineering Pathway places the technical training of the engineer within the broader perspective of a liberal arts education. In this age, engineers cannot always confine themselves to the technical aspects of engineering projects. The modern engineer must be aware of the social responsibilities of the profession, and must be able to explain the possibilities and the risks of new technologies to others. Indeed, many engineering schools have chosen to place added emphasis on the arts and humanities. Choosing

Dual Degree Engineering Program

Southwestern’s dual degree option allows students to spend three years at Southwestern, building a strong foundation in physics, math, and the liberal arts, followed by several semesters at an ABET-accredited engineering school. Students who complete both portions of the program earn:

- A Bachelor of Arts in Applied Physics from Southwestern University, and
- An undergraduate degree in Engineering from the partner engineering school.

We have a formal partnership with Washington University in St. Louis, where students can earn undergraduate degrees from both institutions. Washington University also offers a fast-track option to earn a Master’s in Engineering.

Students may also choose to complete the engineering portion at another ABET-accredited institution of their choice.

Master of Science in Mechanical Engineering

For students interested in pursuing an advanced degree in mechanical engineering, our new partnership with Texas Tech University offers a streamlined path to a master’s degree. Students who complete both portions of the program earn:

- A Bachelor of Arts in Applied Physics from Southwestern, and
- A Master of Science in Mechanical Engineering at Texas Tech.

This program offers an accelerated, cost-effective route to a graduate degree while retaining the depth and breadth of a Southwestern education.

Physics-to-Engineering Graduate Pathway

Some students choose to complete a four-year degree in Physics at Southwestern and then apply to a graduate program in engineering. This option can be especially valuable for those interested in specialized fields like aerospace, biomedical, environmental engineering, etc.

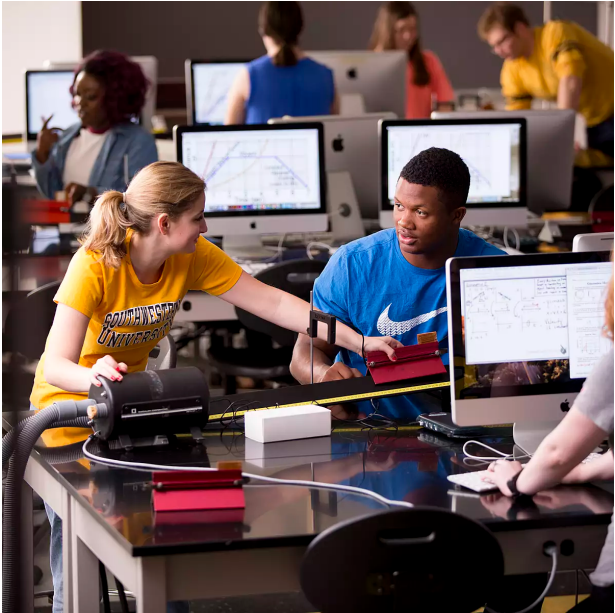
A Bachelor of Arts or Bachelor of Science in Physics from Southwestern provides:

- A solid foundation in mathematics and the physical sciences
- Strong analytical, problem-solving, and communication skills
- The flexibility to pursue a wide range of engineering disciplines in graduate school
- The opportunity to do research with Southwestern faculty

This path allows students to experience the full benefits of a liberal arts education while preparing for advanced study in engineering.

Contact Professor of Physics [Steve Alexander](#) to learn more.

SEE FULL LIST



Southwestern University Partners with Texas Tech University to Create Accelerated Pathway for Baccalaureate and Master’s Degrees

Southwestern and Texas Tech are teaming up to provide Applied Physics majors from SU with the opportunity to pursue a master’s degree in Mechanical Engineering at TTU.

READ FULL STORY

GEORGETOWN, TEXAS

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CONTACT